

PROJECT REPORT BY KUWAR ABHINAV AMAN

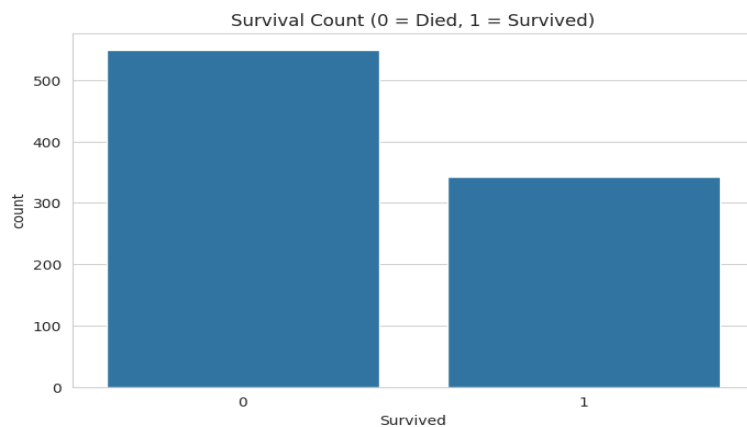
OBSERVATION FOR ALL VISUALS

1. Basic Distributions

1.1 Survival distribution

Observation:

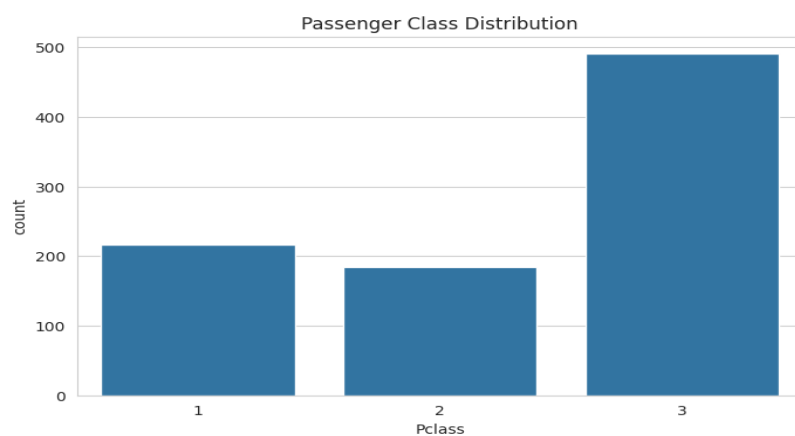
- Only about 38% of passengers survived (342 out of 891)
- Significant class imbalance with more deaths than survivors



1.2 Passenger class distribution

Observation:

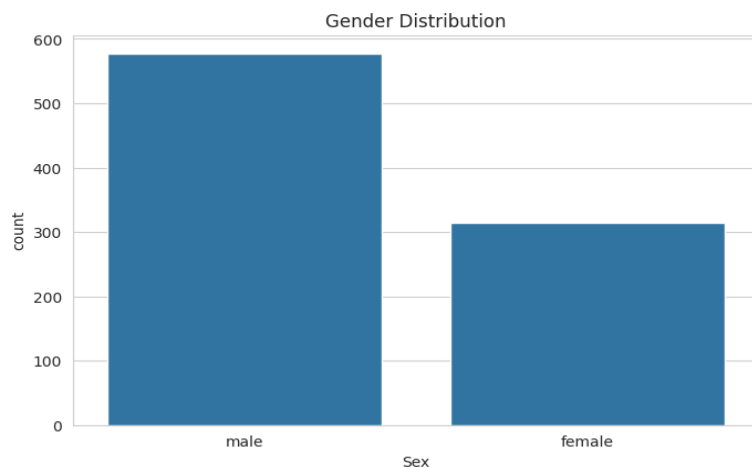
- Majority of passengers were in 3rd class (491 passengers)
- 1st class had 216 passengers, 2nd class had 184



1.3 Gender distribution

Observation:

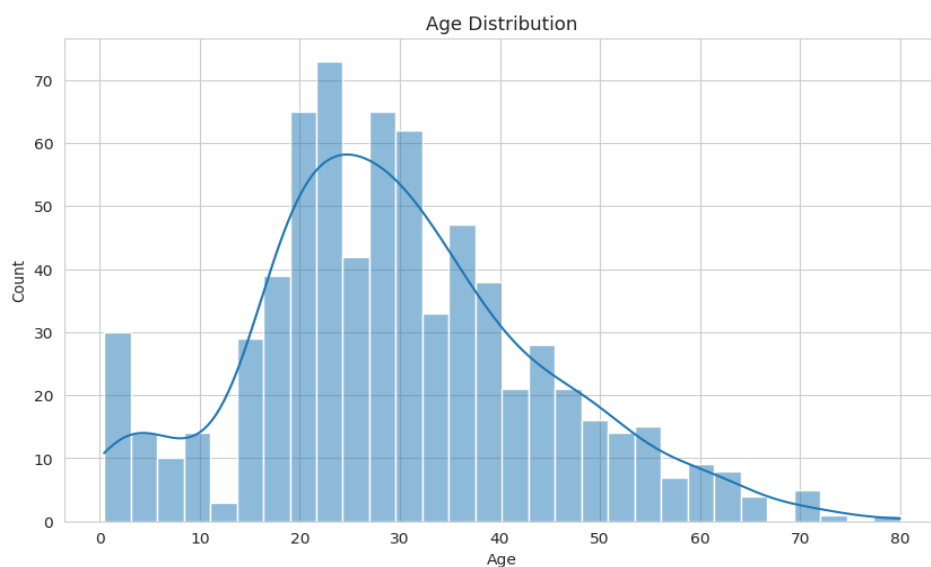
- Significant male majority (577 males vs 314 females)
- About 65% of passengers were male



1.4 Age distribution

Observation:

- Right-skewed distribution with most passengers between 20-40 years
- Few children (under 12) and elderly (over 60)
- Mean age around 30 years

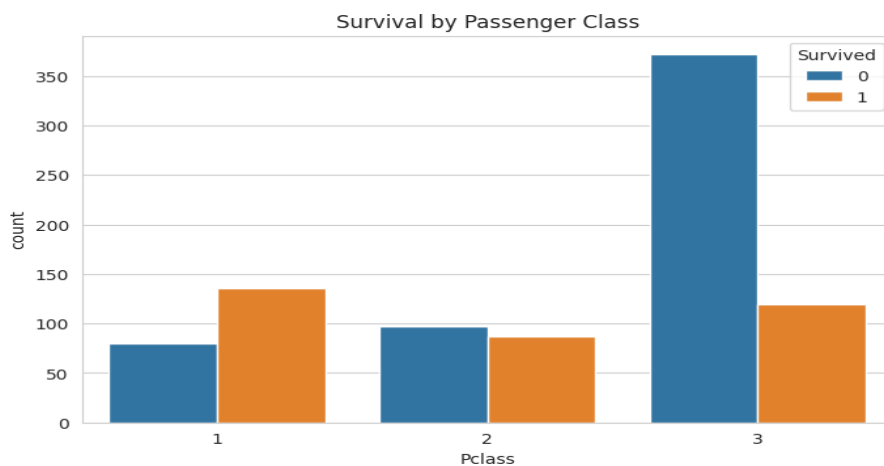


2. Survival Analysis

2.1 Survival by passenger class

Observation:

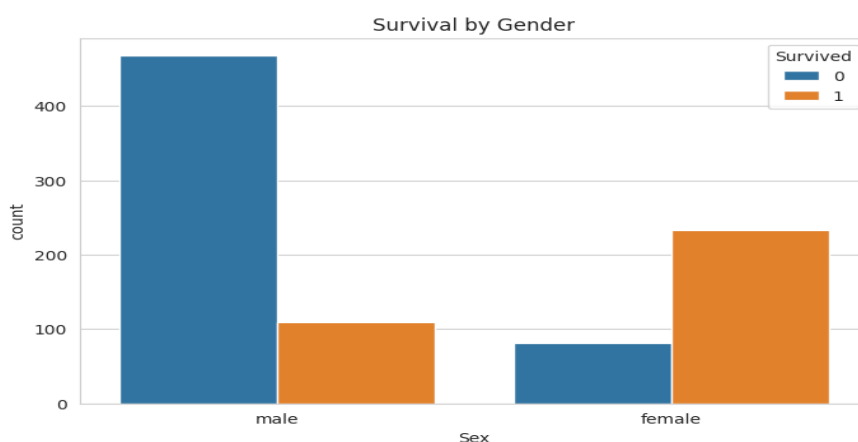
- Strong class bias in survival rates
- 1st class had highest survival rate (~63%)
- 3rd class had lowest survival rate (~24%)



2.2 Survival by gender

Observation:

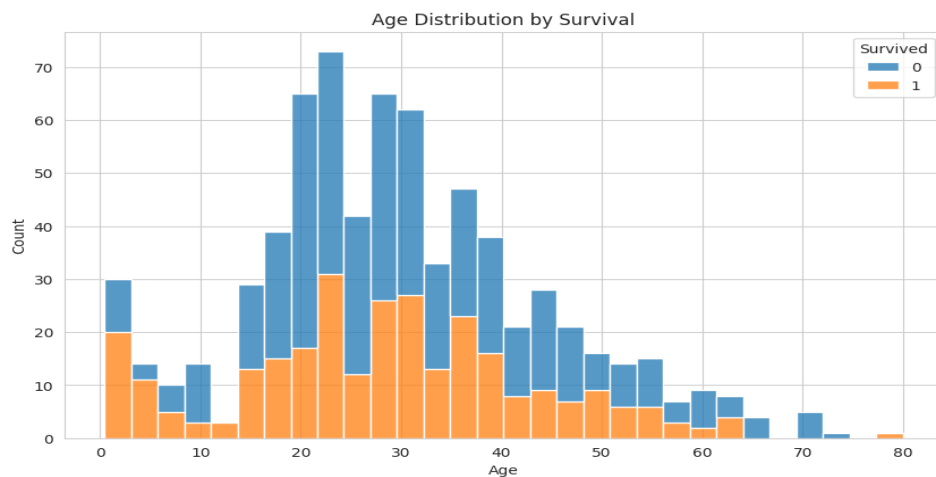
- Dramatic gender difference in survival
- ~74% of females survived vs only ~19% of males
- "Women and children first" policy evident



2.3 Age distribution by survival

Observation:

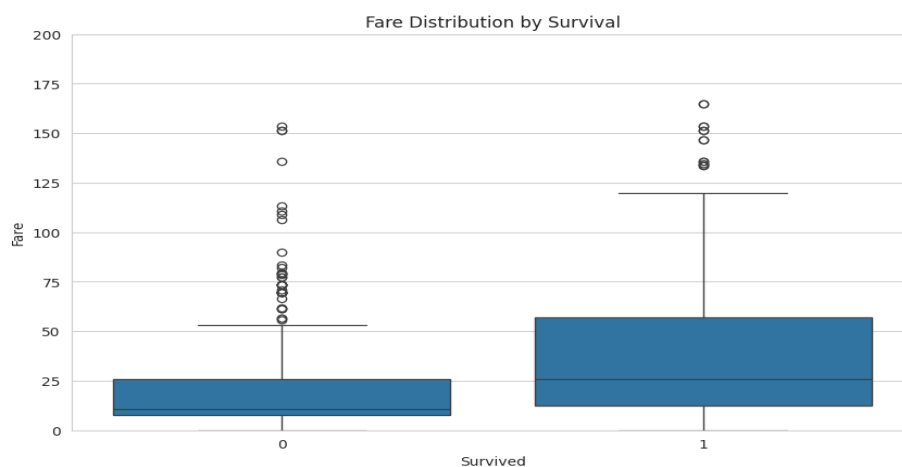
- Children (under 12) had higher survival rates
- Peak deaths in 20-40 age range (most common age group)
- Elderly passengers (60+) had low survival rates



2.4 Fare distribution by survival

Observation:

- Survivors generally paid higher fares
- Median fare for survivors: ~£26 vs ~£8 for non-survivors
- Many outliers in fare amounts, especially among survivors

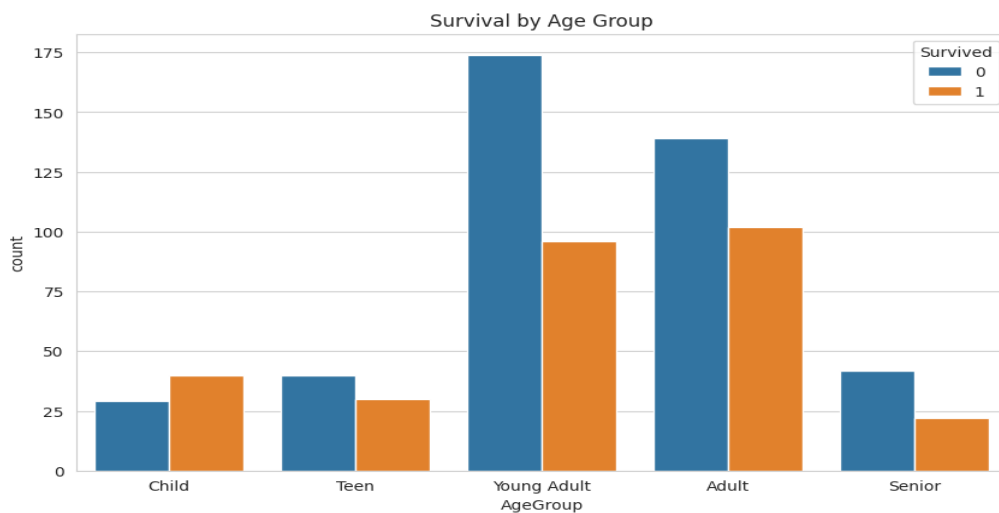


3. Advanced Analysis

3.1 Age groups analysis

Observation:

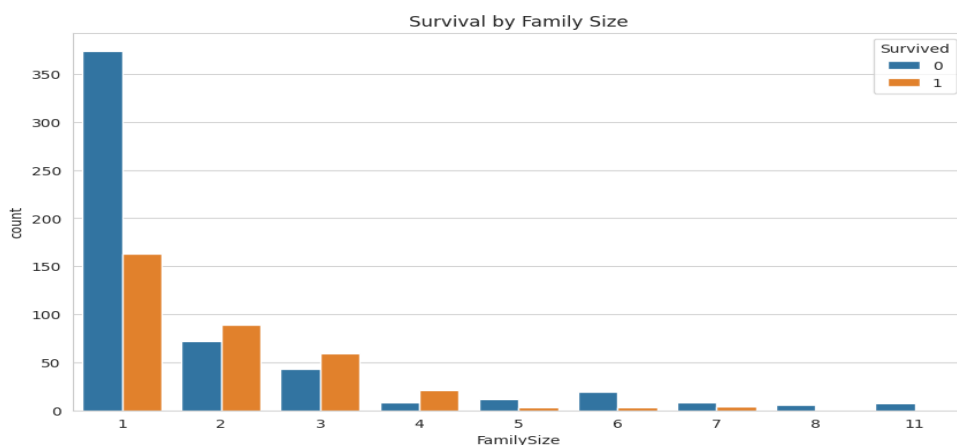
- Children had highest survival rate (~60%)
- Survival rates decrease with age after young adulthood
- Teens and young adults had similar survival rates (~35-40%)



3.2 Family size analysis

Observation:

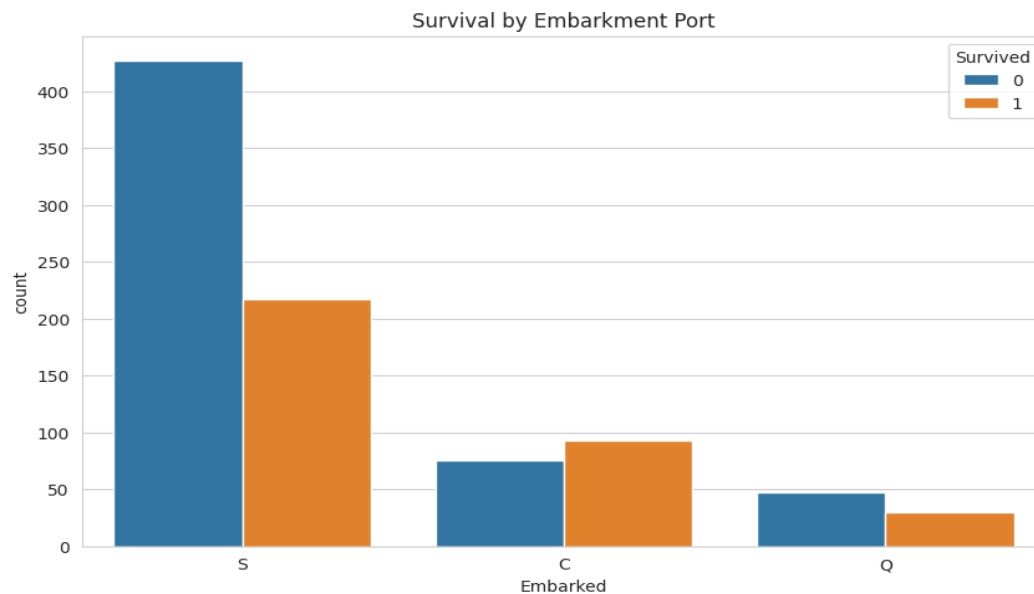
- Solo travelers had lowest survival rate (~30%)
- Families of 2-4 had best survival chances (~50-70%)
- Very large families (5+) had poor survival rates



3.3 Embarkment port analysis

Observation:

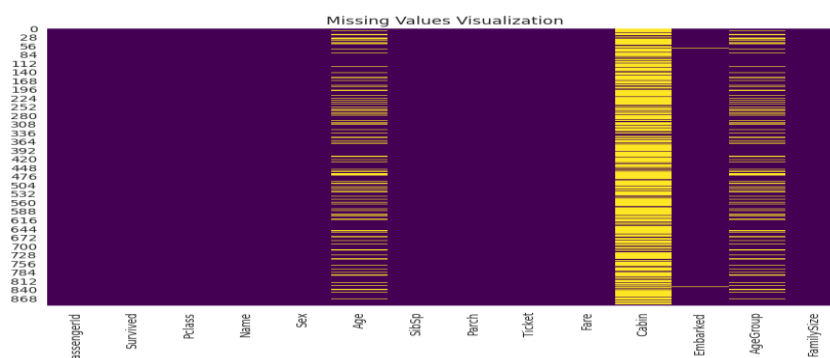
- Cherbourg (C) passengers had highest survival rate (~55%)
- Southampton (S) had lowest survival rate (~34%)
- Possible correlation with passenger class distribution by port



3.4 Missing values visualization

Observation:

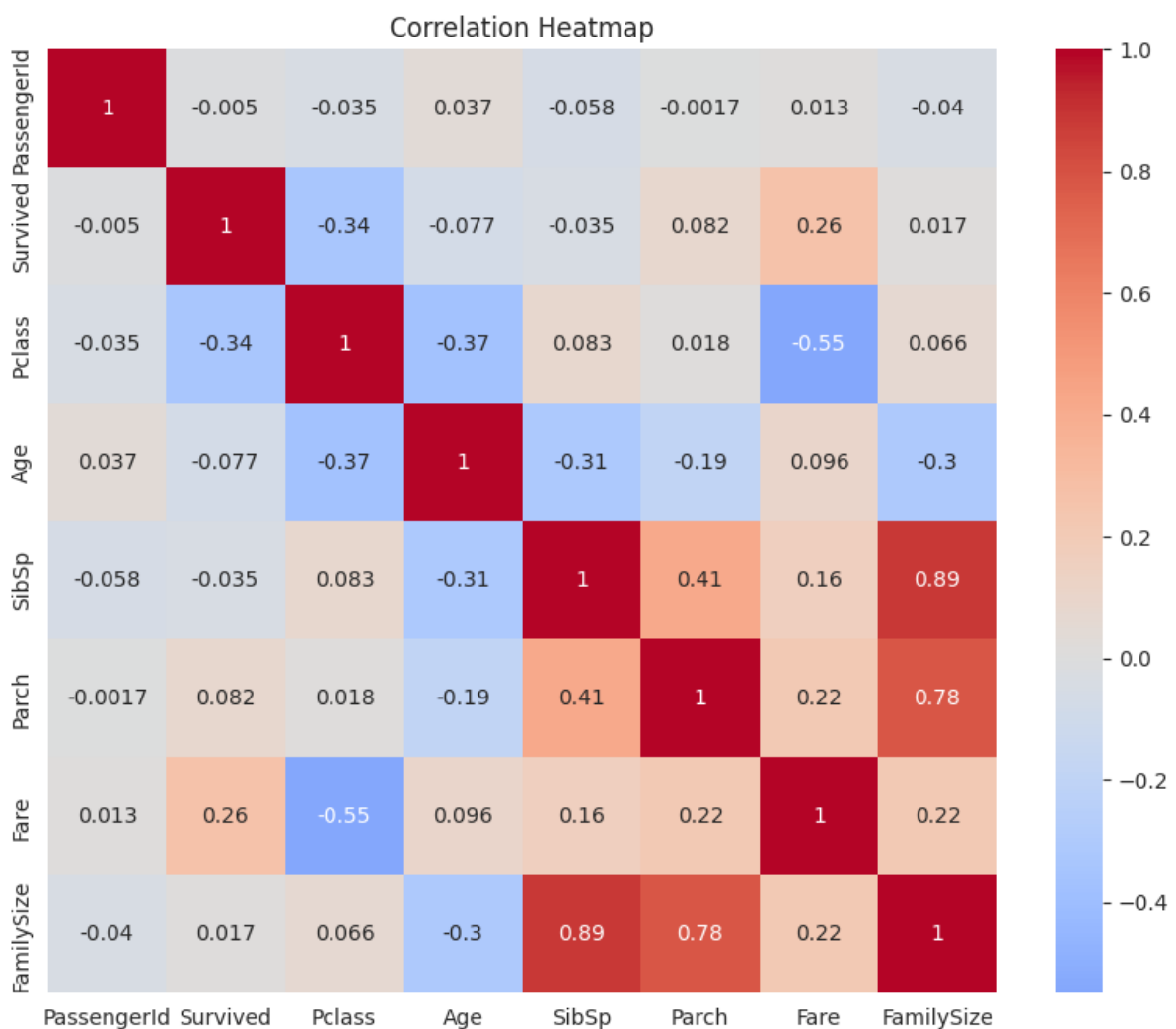
- Age has about 20% missing values (needs imputation)
- Cabin has 77% missing (likely indicates most passengers didn't have cabins)
- Embarked has only 2 missing values



3.5 Correlation heatmap

Observation:

- Strongest positive correlation with survival: Fare (0.26)
- Strongest negative correlation: Pclass (-0.34)
- Age has weak correlation with survival (-0.08)
- Family size (SibSp + Parch) shows small positive correlation (0.02)



FINAL SUMMARY OF FINDINGS:

1. DEMOGRAPHICS:

- 891 passengers in dataset (38% survived)
- Majority male (65%), mostly aged 20-40
- Most common class was 3rd (55% of passengers)

2. SURVIVAL FACTORS:

- Class: 1st class had 63% survival vs 24% in 3rd class
- Gender: 74% female survival vs 19% male
- Age: Children had highest survival rates (60%)
- Fare: Survivors paid significantly higher fares

3. FAMILY IMPACT:

- Solo travellers had worst survival rates
- Small families (2-4 members) had best chances
- Large families struggled to survive together

4. OTHER PATTERNS:

- Cherbourg passengers survived more (55%)
- Cabin information mostly missing (77%)
- Strong correlation between class and survival

5. ANOMALIES:

- Some high-fare passengers in 3rd class
- Few elderly survivors despite "women and children first"
- Several young adult males survived against odds

RECOMMENDATIONS FOR FURTHER ANALYSIS:

1. Investigate interaction effects (e.g., class + gender)
2. Explore title extraction from names (Mr/Mrs/Miss)
3. Examine cabin letter patterns for survivors
4. Model building with these key features